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(1) Research Program/Project/Study Title : Utilization of Snake Plant (<i>Dracaena trifasciata</i>) in Making Textile	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
Municipality/City:		Sigma							
Province:		Capiz							
Region:		VI							
(5) Classification <table border="0"> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☒ Basic</td> <td>☒ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☒ Basic	☒ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
☒ Basic	☒ Pilot Testing								
☒ Applied	☐ Tech. Promotion/Commercialization								
(7) Sector/Commodity		(8) Discipline/Field of Specialization Garments and Fashion Design							

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(9) Rationale

Fibers make an excellent basis for textiles and textile fibers may be created from many natural sources especially from leaves, as well as semi-synthetic methods that use naturally occurring polymers, and synthetic methods that use polymer-based materials, and even minerals such as metals to make foils and wires. Due to the increase in consumption and awareness of environmental impacts, switching to new business models has become mandatory in the textile industry. The researcher noticed that there are many useful resources found and can be utilized into useful products. One thing that caught the attention of the researcher is the Snake Plant (*Dracaena trifasciata*), which grows in open fields and gardens.

Dracaena trifasciata, also known as the snake plant, is a species of flowering plant in the family Asparagus. It is native to tropical West Africa, specifically from Nigeria to the Congo. The plant is most commonly known as the snake plant, Saint George's sword, mother-in-law's tongue, and viper's bowstring hemp, among other names. The snake plant is an evergreen, perennial tropical plant that grows in sunny and warm locations. It has stiff, vertical leaves that give the plant its unique appearance. The leaves grow from the base of the plant and can grow up to 4 feet long and 2-3 inches wide.

Snake Plant (*Dracaena trifasciata*) is the source or material used for making sleeping mats, slippers, and bags. The researcher think a way to produce a smooth and durable textile that can be utilized into useful material, like table mats and costumes.

This study attempt to promote environmentally-friendly and affordable material by substituting common materials to make into something new. Snake Plant (*Dracaena trifasciata*) is abundant in locality as ornamental or decorative plant, and it can particularly thrive anywhere.

Maximizing the potentials of Snake Plant (*Dracaena trifasciata*) may lessen the expenses in the manufacturing of textile, thus, helping the textile industry. It is believed that the source of petroleum-based products are limited and uncertain. Thus, an alternative but cheap sustainable and easily available raw material may be considered. Seeing the potential of Snake Plant (*Dracaena trifasciata*) as a material in textile making, the researcher will experiment in making one out of Snake Plant (*Dracaena trifasciata*). This study needs to characterize whether it can produce a useful product for the benefits of the people.

(9-A) Project Description

Textile from Snake plant is used in making produce a smooth and durable textile that can be utilized into useful material, like table mats and costumes.

(10) Objectives/Statement of the Problem

This study aims to develop a textile from Snake Plant (*Dracaena trifasciata*). Specifically to:

- 1.) determine the qualities of Snake Plant (*Dracaena trifasciata*) textile in terms of reaction to acid and base, burning, tensile strength, water absorption, and expansion and contraction of fiber.
2. determine the level of acceptability of Snake Plant (*Dracaena trifasciata*) textile in terms of feeling, presence of plugs, irregularity of the ribs, color shade, and color variation.

(11) Review of Literature/Prior Art Search

Natural fibers present important advantages such as low density, appropriate stiffness and mechanical properties and high disposability and renewability. (Thus, Hemp (*Cannabis sativa*) fibers and Snake Plant (*Dracaena trifasciata*) fibers are belong to natural fibers and possesses these qualities.

According to the study of (Balter, 2019).Fibers are of two types: Natural fiber, which consists of animal and plant fibers, and man-made fiber, which consists of synthetic fibers and regenerated fibers. After World War II, the build-up of synthetic fibers significantly decreased the use of natural fibers. Now, with the increase of oil prices and environmental considerations, there has been a revival of natural fiber use within the textile, building, plastic, and automotive industries. This interest is reinforced by the developmental perspectives on the agro-industrial market and local productions, allowing economic development and independence versus imported materials (Balter, 2019)

Natural fibers are made from plant, animal and mineral sources. Natural fibers can be classified according to their origin.

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Table 1. Category of Natural Fibers

Category	Description
Seed Fiber	Fibers collected from seeds or seed cases. e.g. cotton and kapok
Leaf Fiber	Fibers collected from leaves. e.g., sansevieria, fique, sisal, banana and agave.
Bast Fiber	Fibers are collected from the skin or bast surrounding the stem of their respective plant. These fibers have higher tensile strength than other fibers. Therefore, these fibers are used for durable yarn, fabric, packaging, and paper. Some examples are flax, jute, kenaf, industrial hemp, ramie, rattan, and vine fibers.
Skin Fiber	
Fruit Fiber	Fibers are collected from the fruit of the plant, e.g. coconut (coir) fiber.
Stalk Fiber	Fibers are actually the stalks of the plant. E.g. straws of wheat, rice, barley, and other crops including bamboo and grass. Tree wood is also such a fiber.

* Balter M. (2019). Clothes Make the (Hu) Man Retrieved January 30, 2024



Figure 1. General views of the Hemp Plant (*Cannabis sativa*). (The gardens and the Trieux Labyrinth of Hemp work by Barbetorte August 10, 2019)

Hemp fiber has been used extensively throughout history, with production climaxing soon after being introduced to the New World. Items ranging from rope, to fabrics, to industrial materials were made from hemp fiber. Hemp is often used to make sail canvas, and the word canvas derives from cannabis. Today, a modest hemp fabric industry exists, and hemp fibers can be used in clothing. Pure hemp has a texture similar to linen (Cronin, 2015).

Traditionally the hemp stalks would be water-retted first before the fibers were beaten off the inner hurd by hand; a process known as scrutching. As mechanical technology evolved, separating the fiber from the core is accomplished by crushing rollers and brush rollers that would produce a nearly clean fiber. After the Marijuana Tax Act is implemented in 1938, the technology for separating the fibers from the core remained "frozen in time".

Only in 1997, starting with Ireland, did the Commonwealth countries and then other countries begin to legally grow industrial hemp again. Reiterations of the 1930's "decorticator" have been met with limited success, along with steam explosion and chemical processing known as thermomechanical pulping.

Mechanical harvesting is now common, using specially adapted cutter-binders or simpler cutters. The cut hemp is laid in swathes to dry for up to four days. This is traditionally followed by retting, either water retting (the bundled hemp floats in water) or dew retting (the hemp remains on the ground and is affected by the moisture in dew, and by molds and bacterial action) (Statistical and Economic Information, 2021).

Hemp benefits crops grown after it. For this reason, it is generally grown before winter cereals. Advantageous changes are high weed suppression, soil loosening by the large hemp root system and the positive effect on soil tilth. Since hemp is very self-compatible, it can also be grown several years in a row in the same fields (monoculture) (Thomas, 2019).

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Hemp is considered by a 1998 study in Environmental Economics to be environmentally friendly due to a decrease of land use and other environmental impacts, indicating a possible decrease of ecological footprint in a US context compared to typical benchmarks. A 2010 study, however, that compared the production of paper specifically from hemp and eucalyptus concluded that "industrial hemp presents higher environmental impacts than eucalyptus paper"; however, the article also highlights that "there is scope for improving industrial hemp paper production". Hemp is also claimed to require few pesticides and no herbicides, and it has been called a carbon negative raw material. Results indicate that high yield of hemp may require high total nutrient levels (field plus fertilizer nutrients) similar to a high yielding wheat crop (Risula, 2019).

Fibers and Textiles from Plants

A variety of plants are sources of natural fibers, as discussed in the following: Stems of milkweed plants have been used to obtain natural cellulose fibers with better strength and elongation than the milkweed floss fibers. Milkweed stem fibers have high cellulose content but low % crystallinity. The fibers have strength similar to cotton and elongation higher than that of linen fibers. The modulus and moisture regain of the milkweed stem fibers is between that of cotton and linen. Overall, the milkweed stem fibers have properties required for high value textile, composite, and other fibrous applications. Utilizing the milkweed stems for high quality natural cellulose fibers will add value and make milkweed a more useful fiber plant (Reddy and Yang, 2019).

According to the study of (Abdullah 2019) *Sansevieria trifasciata* L. is a wild leafy fibres plant. It is a source of strong white fibre. They are commonly used as rope, fishing lines/net, cordage, bowstring and clothing materials. From the results of the above study and information, it can be inferred that this



Fig. 2. Preliminary observation of STLF and products. (a) Untreated fibres, (b) bleached fibres, (c) dyed fibres, (d) cellulose and (e) crude microcrystalline cellulose.

fibre and other parts of this plant have the potential for wide-scale used and diversified product development for the use as a raw material in different industries such as textile and garments, pulp and paper, pharmaceuticals and cosmetics along with various decorative furnishing industries. Renewable and ecologically natural fibres are in great demand for a sustainable and pollution-free planet. *Sansevieria trifasciata* L. is leafy and perennial fibre. Climatic and agro-ecological conditions reflect the possibility of wide-scale production in our non-agricultural and bushy barrel land along with newly practices urbanized roof-agriculture and environmental mitigation.

Related Literature

Two Chemical Treatment Cellulose micro fibrils were prepared using different combinations of chemical and mechanical treatments. Three different treatments were carried out on different samples while some was left as control. The treatment was carried out as follows: Mass of Fibre used: 120g sisal fibre was divided into four equal mass of 30g each. 2.3.2.1 Sample A. The sample was treated with 0.5M NaOH solution (450ml) inside shaker water bath at 40°C for 10 hours. The insoluble residue was treated with 0.5M NaOH and 3wt% H₂O₂ solution (400ml) at 45°C for 8hours and was finally treated with 2M NaOH solution (400ml) at 55°C for 2hours. 2.3.2.2 Sample B The sample was treated with 0.5M NaOH solution (450ml) inside shaker water bath at 40°C for 10 hours. The insoluble residue was treated with 0.5M NaOH and 3wt% H₂O₂ solution (400ml) at 45°C for 8hours and finally treated with 2M HCl (400ml) at 70°C for 2hours in order to remove mineral traces. 2.3.2.3 Sample C This sample was treated with 5wt% KOH solution (450ml) inside shaker water bath at 40°C for 10hours. The insoluble residue was delignified with 1% NaCl solution (400ml) at pH 3, and adjusted with 10wt% acetic acid at 70°C for 1hour. Another

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treatment with KOH solution (250ml) under the same conditions as the first step was used. Finally, a 1wt% HCl solution (300ml) at 80°C for 2hours was used to remove mineral traces.

From the results, it was observed that the lignin and hemicelluloses constituents as well as their pH values have the same response to the chemical treatments. The results show higher percentages of lignin and hemicelluloses in sample A that was treated with alkaline peroxide more than that of the control sample while samples B treated with HCl in addition to the alkaline peroxide and C treated with HCl in addition to KOH, acetic acid and NaCl have lower values. However, it was observed that sample B has the least values for both constituents. This result was in agreement with the results from the above mentioned researchers. (wolesuccess2000@yahoo.com)

Cotton, linen, and other plant fibers are made mostly of cellulose, which has the ring-like structure. When cellulose burns, it takes in oxygen and puts out carbon dioxide and water. As a balanced chemical equation: $C_6H_{10}O_5 + 6O_2 \xrightarrow{\text{heat}} 6CO_2 + 5H_2O$. The output is gas, so they just float away. Ash that remains from burning anything is mostly made up of metal oxides. (www.absonant.com)

Based on the study of (Singha, 2018) on **Tensile strength test**: The tensile strength test was conducted on Computerized Universal Testing Machine (HOUNSFIELD H25KS). The sample of 10 cm length was clamped into the two jaws of the machine. Each end of the jaws covered 2 cm of the sample. Tensile strength was studied over the rest of 6 cm gauze length. Reading of the tensile strength test instrument for Newton force and extension was initially set at zero. The test was conducted at the constant strain rate of the order of 10 mm/min. Tensile stress was applied till the failure of the sample and load-extension curve was obtained. Each sample was tested five times and average results have been reported (Singha, 2018).

Natural Fibers

Natural fibers are having low cost, low density and are biodegradable. All these points are on the positive side but they are hydrophilic. Their water absorption property limits the use of these fibers as potential reinforcement in the preparation of composites. Polymer matrix is hydrophobic in nature. This makes fiber and matrix incompatible and results in poor interfacial bonding between the fiber and the matrix. The main objective of this chemical treatment is to reduce their water absorption property and also to improve the compatibility with polymer matrix. Abaca/Polypropylene composites with 20-50 wt% fiber loading have been developed by hot compression molding technique and were then analyzed for tensile strength. The study revealed that the tensile strength of abaca composites increased upon alkali and benzenediazonium chloride treatment.

Table 2. Chemical Composition of Abaca Fiber:

Parameter	Percentage (%)
Cellulose	76.6%
Hemicellulose	14.6%
Lignin	8.4%
Pectin	0.3%
Wax and fat	0.1%

(http://fida.da.gov.ph/Templates/abaca_grades_and_uses.htm)

The results show that tensile strength of the composites increased with increase in fiber loading up to 40% and beyond 40% it showed a decrease in tensile strength. From the observations in this study we can say that chemically treated abaca fibers can be used as promising materials as reinforcement in the preparation of bio composites (Punyamurthy, 2014).

According to the study of Geronaga (2018) on Feasibility of Hard Fibers from Sansevieria Trifasciata (Snake Plant) as Fabric: The extraction of natural fibres from plants and their uses for the production of fabrics and other woven or constructed textile materials has played a major role in textile development today. Thus, I investigated the potential of snake plant to produce natural fibres as a fabric, which would be acceptable in terms of its moisture regain, absorbency, crimp frequency, and linear density. Dried fibres from snake plant were subjected for fibre extraction using fibre-decorticating machine. The extracted fibres were then washed thoroughly and was wrung to remove excess liquids and

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other residues. The fibres were then sun-dried and were combed to remove any residues. The fibres had been chemo-mechanically softened, and were later spun using a spinning machine. The yarn was weaved using the weaving machine, and the fabric was then tested for moisture regain, absorbency, crimp frequency, and a linear density often replications. Results showed that fibres from *Sansevieria trifasciata* have no crimps with a recorded mean average of 0.1 crimps per fibre and 0.483 grams per cubic meter in linear density. Moisture regain of 8.5% and a mean average of 40.8 seconds was recorded for the absorbency respectively. This study indicated that *Sansevieria trifasciata* (Snake plant) is capable of producing fabric out of its hard fibres. It also proves that the soft and tough nature of the natural fibres can never be equated to the hard and coarse nature of the human-made nylon fibre especially when they are handled in tying or knotting of any type.

According to the study of Li 2023 in order to study the possibility of flexible textile composites as space deployable devices, carbon fiber triaxial woven fabric and thermoplastic polyurethane were compounded by hot-pressing. The section shape of the carbon fiber triaxial woven fabric flexible textile composites was observed by scanning electron microscopy. After hot-pressing at 120°C for 45 min, the thermoplastic polyurethane is best combined with carbon fiber bundle, and fully melted to form a protective layer, which also ensured the deformability of carbon fiber triaxial woven fabric/thermoplastic polyurethane flexible textile composites. The tensile strains of flexible textile composites in the directions of 0°, 15° and 30° were analyzed by digital image correlation. The analysis was carried out from the macroscopic scale, the single cell scale, and the fiber bundle scale, combined with the strain distribution of digital image correlation in the tensile process. From the axial and radial deformation of the bundle and the force analysis of single cell, the phenomena such as strain change, trellising phenomenon, and fabric deformation during tensile were explained. When carbon fiber triaxial woven fabric is added to glass-fiber composite, the electromagnetic reflection efficiency can be increased by 9%. The research on strain and electromagnetic reflection provided the possibility for the flexible textile composites as a space deployment device.

The study of **Apalla 2021)** aimed to investigate the potential of *Sansevieria Trifasciata* fiber for textile application. By using the *Sansevieria Trifasciata* fiber, the researcher attempted to produce a fabric that can compete with other native fabrics here in the Philippines. This natural fiber from *Sansevieria Trifasciata* is hoped to support, promote and encourage the growth and development of other native products in the Philippines particularly in the province of Isabela. To assess the acceptability of *Sansevieria Trifasciata* fabric, a group of students and teachers/trainers were asked to do the evaluation along these criteria: appearance, durability, usefulness and eco-friendliness. Results of evaluation showed high acceptability of the product.

The product has unique property which makes it capable of withstanding pressure without any fiber degradation. The developed fabric's breaking strength likewise revealed that it has physical potential to be used in manufacturing goods made of fabric. Due to its great strength, cost-effectiveness and renewable source, fabric developed from *Sansevieria Trifasciata* fiber has potential use in the clothing industry.

Snake Plant (*Dracaena trifasciata*)

Dracaena trifasciata is a species of flowering plant in the family Asparagusaceae, native to tropical West Africa from Nigeria east to the Congo. It is most commonly known as the snake plant, Saint George's sword, mother-in-law's tongue, and viper's bowstring hemp, among other names. Until 2017, it was known under the synonym *Sansevieria trifasciata* (Roscoe, 2021).



Figure 2. Real picture of snake plant as a common ornamental plant that can be found everywhere.

It is an evergreen perennial plant forming dense strands, spreading by way of its creeping rhizome, which is sometimes above ground, sometimes underground. Its stiff leaves grow vertically from a

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basal rosette. Mature leaves are dark green with light gray-green cross-banding and usually range from 70–90 centimetres (2.3–3.0 ft) long and 5–6 centimetres (2.0–2.4 in) wide, though it can reach heights above 2 m (6 ft) in optimal conditions (Harrison, 2021).

The plant exchanges oxygen and carbon dioxide using the crassulacean acid metabolism process, which allows them to withstand drought. The microscopic pores on the plant's leaves, called the stomata and used to exchange gases, are opened only at night to prevent water from escaping via evaporation in the hot sun. This plant is often kept as a houseplant due to its non-demanding maintenance; they can survive with very little water and sun. To get this plant to go into bloom outside of its natural environment is difficult. Replicating its natural environment is possible. Its flowers vary from greenish white to cream-colored — some are fragrant at night, others not at all — and have a sticky texture. *Dracaena trifasciata* is commonly called "mother-in-law's tongue", "Saint George's sword" or "snake plant", because of the shape and sharp margins of its leaves that resemble snakes. It is also known as the "viper's bowstring hemp", because it is one of the sources for plant fibers used to make bowstrings United State Department of Agriculture, 2015).

Like some other members of its genus, *D. trifasciata* yields bowstring hemp, a strong plant fiber once used to make bowstrings. It is now used predominantly as an ornamental plant, outdoors in warmer climates, and indoors as a houseplant in cooler climates. It is popular as a houseplant because it is tolerant of low light levels and irregular watering; during winter, it needs only one watering every couple of months. It will rot easily if overwatered. It is commonly recommended to beginners interested in cultivating houseplants for its easy care (Roscoe, 2021).

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Method

This study is an experimental method of research wherein it aims to make a textile out of Snake Plant (*Dracaena trifasciata*) utilizing to a textile through loom weaving. This experimentation is the possibility that Snake Plant (*Dracaena trifasciata*) can be made into cloth.

The purpose of this experimental study is to make an alternative if the raw materials is not available like for example piña cloth, abaca cloth, banana cloth, and clothing and textiles have been important in human history and reflects the materials available to a civilization as well as the technologies that had been mastered. The social significance of the finish product reflects their culture.

Sources available for the study of clothing and textiles include material remains discovered via archaeology; representation of textiles and their manufacture in art; and documents concerning the manufacture, acquisition, use, and trade of fabrics, tools, and finished garments. Scholarship of textile history, especially its earlier stages, is part of material culture studies.

Materials

The materials to be used in the study is Snake Plant (*Dracaena trifasciata*) leaves and then the fiber will be extracted out of it. The tools in gathering and extracting fibers are bolo and spoon and the equipment to be used in weaving of fiber are the Handloom, pot with sand, winder straw and shuttle. The tools and materials to be use in testing the devices are hydrochloric acid (HCl), Potassium Hydroxide (KOH), water, spring balance, graduated cylinder, beaker, thong, alcohol lamp, thermometer, masking tape, matches, pencil, tweezers, Vernier caliper, analytical balance, pentel pen, tissue paper.

Design Plan

An experimental design/layout will be follow to guide the researcher in the preparation, development as well as in the evaluation of the product. Steps and procedures are observe and noted as to whether unanticipated events or changes had occur. A detailed procedure or steps will be formulated. It will reflects the flow of work from gathering of needed materials, preparation, and development up to the product is finished for clarity and understanding.

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Procedure of the Study

General procedure or steps in the production of Snake Plant (*Dracaena trifasciata*) Textile first is to gather all the materials needed in the production of Snake Plant (*Dracaena trifasciata*) Textile. Then after the leaves are gathered the fiber is then extract as it is more advantageous if the fiber is extracted when the leaves are still fresh as it is harder if the leaves are dried. When extracting the fiber, the leaves are pounded using rounded wood and then the fiber is scraped using metal comb. After the fiber is extracted it is then dried in the sun for several days and after that it will be soaked in a hot water and joint together and tie as knot and put into a bobbin then process into textile.

Collection of Data

The research instrument use in the study is the evaluation sheet. Computerized Universal Testing Machine (HOUNSFIELD H25KS) will be used to determine the qualities of Snake Plant (*Dracaena trifasciata*) textile in terms of reaction to acid and base, burning, tensile strength, water absorption, and expansion and contraction of fiber.

The mean will be used to determine the level of acceptability of Snake Plant (*Dracaena trifasciata*) textile as a product in terms of feeling, presence of plugs, irregularity of the ribs, color shade, and color variation.

Scoring the Variables

In scoring the variables, the researcher use the Five Points Rating Scale in evaluating the product.

Score	Mean	Verbal Interpretation
5	4.21-5.00	Very Acceptable
4	3.41-4.20	Acceptable
3	2.61-3.40	Moderately Acceptable
2	1.81-2.60	Less Acceptable
1	1.00-1.80	Least Acceptable

1. Feeling

Score	Mean	Verbal Interpretation
5	4.21-5.00	Very Acceptable
4	3.41-4.20	Acceptable
3	2.61-3.40	Moderately Acceptable
2	1.81-2.60	Less Acceptable
1	1.00-1.80	Least Acceptable

2. Presence of plugs

Score	Mean	Verbal Interpretation
5	4.21-5.00	Very Acceptable
4	3.41-4.20	Acceptable
3	2.61-3.40	Moderately Acceptable
2	1.81-2.60	Less Acceptable
1	1.00-1.80	Least Acceptable

3. Irregularity of the ribs

Score	Mean	Verbal Interpretation
5	4.21-5.00	Very Acceptable
4	3.41-4.20	Acceptable
3	2.61-3.40	Moderately Acceptable
2	1.81-2.60	Less Acceptable
1	1.00-1.80	Least Acceptable

4. Color shade

Score	Mean	Verbal Interpretation
5	4.21-5.00	Very Acceptable
4	3.41-4.20	Acceptable
3	2.61-3.40	Moderately Acceptable
2	1.81-2.60	Less Acceptable
1	1.00-1.80	Least Acceptable

5. Color variation

Score	Mean	Verbal Interpretation
5	4.21-5.00	Very Acceptable
4	3.41-4.20	Acceptable

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3	2.61-3.40	Moderately Acceptable
2	1.81-2.60	Less Acceptable
1	1.00-1.80	Least Acceptable

Parameters for Analysis

To ascertain the characteristics and components of Snake Plant (*Dracaena trifasciata*) Textile, the following parameters are consider.

1. The mean formula is (Downie and Heath, 1984)

$$\bar{X} = \frac{\sum X}{N}$$

Where:

$\bar{X}$	= Arithmetic Mean
$\sum X$	= Sum of raw scores
$\bar{N}$	= No. of respondents

2. Burning rate of Snake Plant (*Dracaena trifasciata*) Textile. To compute for the Area of 1X1 dimension of Snake Plant (*Dracaena trifasciata*) Textile;

$$A = L \times W$$

Whereas;

A = Area

L = Length

W = Width

1 inch = 2.54 centimeter

1 centimeter= 10 millimeter

$$27.6^{\text{mm}} \times \frac{1 \text{ cm}}{10 \text{ mm}} \times \frac{1 \text{ in}}{2.54 \text{ cm}} = \frac{2.76 \text{ in}}{2.54 \text{ cm}} = 1.09^{\text{in}^2}$$

$$\text{Burning rate} = \frac{\text{Area}}{\text{Time}}$$

Percentage of water absorption, expansion and contraction of Snake Plant (*Dracaena trifasciata*) textile.

$$\% = \frac{\text{Final}-\text{Initial}}{\text{Final}} \times 100$$

Cost Analysis

Table 3. Cost analysis of materials use in the production of Snake Plant (*Dracaena trifasciata*) Textile.

Quantity	Unit	Description	Value	Remarks
70	Leaves	Snake Plant	₱100.00	
1	pc.	Clay pot	₱ 35.00	
2	cups	sand	₱ 20.00	
TOTAL COST OF MATERIALS			155.00	
COST OF LABOR (40%)			62.00	
TOTAL COST OF PRODUCT			217.00	

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(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Research Objectives	Year 1						Year 2					
	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
1. conceptualization of the proposal												
2. proposal Writing												
3. Proposal presentation												
4. Conduct of the Study												
5. Data gathering												
6. Data analysis												
7. Discussion and Recommendation												

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

Textile from Snake plant is used in making produce a smooth and durable textile that can be utilized into useful material, like table mats and costumes.

(16) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

The Snake Plant (*Dracaena trifasciata*) will be used as an alternative if the raw materials is not available like for example piña cloth, abaca cloth, banana cloth, and clothing and textiles have been important in human history and reflects the materials available to a civilization as well as the technologies that had been mastered. The social significance of the finish product reflects their culture.

(17) Target Beneficiaries (Who the clienteles are and what are the expected outcome/effects of the use of the projects output)

Dress shop owners, Dressmakers/Tailor, Fashion designers, Native people, Textile entrepreneurs, Trade

(18) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(19) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)		1,000.00
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Transportation allowance		500.00
- Meal allowance during data gathering		500.00
- Professional fees (laboratory analysis and other related fees)		3,000.00
- Miscellaneous (for incidental expenses)		2,000.00
- Printing		2,000.00
C. CAPITAL OUTLAY		9,000.00
Total		

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(20) Literature Cited (include only those references published in reputable journals, and at least that from **CAPSU Research Journal**, <http://researchjournal.capsu.edu.ph/>)

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 Cabral (2019) he suggested that the Snake Plant Fiber, which served as alternative matting in making fiberglass, have various uses aside from the focus of this study. Cabral (2019)
 Geronaga (2018) Feasibility of Hard Fibers from Sansevieria Trifasciata (Snake Plant) as Fabric
 (Fidan, 2020)Integrated Life Cycle Assessment Approach for Denim Fabric Production Using Recycled Cotton Fibers and Combined Heat and Power Plant
 (Alelie M. Apalla 2021) *Development of Fabric from Sansevieria Trifasciata (Snake Plant)*
 Abdullah 2020)Extraction and Proximate Study of Sansevieria Trifasciata L. As Fibre Source for Textile and Other Uses, Journal of the Asiatic Society of Bangladesh Science 46(2):155-162

(21) Capsule Curriculum Vitae

Name : **JENNY CALSADO PEREZ**
 Address : **POBLACION ILAYA, DUMALAG, CAPIZ**
 Date of Birth : **APRIL 10,1994**
 Age : **29**

PERSONAL PROFILE

Sex : **FEMALE**
 Civil Status : **SINGLE**
 Height : **5'3"**
 Weight : **61KG.**
 Citizenship : **Filipino**
 Religion : **ROMAN CATHOLIC**
 Father's Name : **LUCIANO F. PEREZ**
 Mother's Name : **ROQUITA C. PEREZ**



EDUCATIONAL BACKGROUND

Graduate Studies:	Doctor of Education Major in Industrial Education Fuentes Drive, Roxas City, Capiz	2023-Present
Graduate studies :	Master of Arts in Industrial Education Major in Home Economics Fuentes Drive, Roxas City, Capiz	2020-2023
Tertiary	Bachelor of Secondary Education Major in Technology and Livelihood Education Fuentes Drive, Roxas City, Capiz	2015-2019
Secondary :	Dumalag Central National High School Poblacion Ilaya, Dumalag, Capiz	2007-2011
Elementary :	Dumalag Central School Dumalag, Capiz	2000-2006

ELIGIBILITY

Passed the Licensure Examination for Teachers with a rating of 80.40. **2019**

TRAININGS

Attended training and passed the National Competency Assessment in:

CONSTRUCTION PAINTING NCII	November 2020
DRESSMAKING NCII	November2020-January 2021
TRAINER'S METHODOLOGY NCI	September –November 2021